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Predicting the Wildland Fire Spread Using a Mixed-Input CNN Model with Both Channel and Spatial Attention Mechanisms

Xingdong Li · Xinyu Wang · Shufa Sun · Yang Wang · Sanping Li · Dandan Li

Environmental Science, Computer Science · Fire technology · 20 June 2023

TLDR The proposed fire-**spread**ing model MI-AM-CNN using a convolutional neural network with mixed-inputs and attention mechanisms has the highest performance in predicting effectiveness and efficiency and its predicted results can provide theoretical support for the forest fire **spread prediction** method based on artificial intelligence. Expand

12 · Springer Nature · Save · Cite

Fusion of Physics-Based Wildfire Spread Models with Satellite Data using Generative Algorithms

B. Shaddy · Deep Ray · +7 authors · A. Oberai · Environmental Science, Physics

TLDR A method is developed to infer the history of a **wildfire** from satellite measurements using a conditional Wasserstein Generative Adversarial Network (cWGAN), providing the information necessary to initialize coupled atmosphere-**wildfire** models in a physics-informed approach based on measurements. Expand

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Assessment of Wildfire Susceptibility and Wildfire Threats to Ecological Environment and Urban Development Based on GIS and Multi-Source Data: A Case Study of Guilin, China

Weiting Yue · C. Ren · +4 authors · Zhenkui Wei · Environmental Science, Geography · Remote Sensing · 19 May 2023

The frequent occurrence and **spread** of **wildfires** pose a serious threat to the ecological environment and urban development. Therefore, assessing regional **wildfire** susceptibility is crucial for the… Expand

32 · PDF · Publisher · Save · Cite

Operational wildfire suppression modelling: a review evaluating development, state of the art and future directions

T. Duff · K. Tolhurst · Environmental Science · 21 September 2015

Wildfires are an inherent part of the landscape in many parts of the world; however, they often impose substantial economic burdens on human populations where they occur, both in terms of impacts and… Expand

66 · Publisher · Save · Cite

Data-driven spatio-temporal analysis of wildfire risk to power systems operation

A. Umunnakwe · M. Parvania · Hieu Nguyen · J. Horel · K. Davis

Environmental Science, Engineering · IET Generation, Transmission & Distribution · 13 May 2022

TLDR The proposed model, based on real-world dataset, is demonstrated on the IEEE 24-bus test system mapped to a study area in northern California, and results illustrate the proposed model can achieve the best performance in potential wildfire ignition detection. Expand

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Radiomics-based prediction for tumour spread through air spaces in stage I lung adenocarcinoma using machine learning.

Donglai Chen · Y. She · +7 authors · Chang Chen · Medicine ·

European Journal of Cardio-Thoracic Surgery · 3 February 2020

OBJECTIVES As evidence has proven that sublobar resection is oncologically contraindicated by tumour **spread** through air spaces (STAS), its preoperative recognition is vital in customizing surgical… Expand

46 PDF PubMed Save Cite

A review of train delay prediction approaches

Thomas Spanninger · Alessio Trivella · Beda Büchel · F. Corman · Engineering ·

Journal of Rail Transport Planning & Management · 1 June 2022

TLDR It is found that data-driven approaches might have the edge on event-driven approach in terms of **prediction** accuracy, whereas event- driven approaches that explicitly model the dynamics and dependencies of railway traffic have their strength in providing interpretable **predictions**, and are more robust concerning disruption scenarios. Expand

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Data Injection at Execution Time in Grid Environments Using Dynamic Data Driven Application System for Wildland Fire Spread Prediction

Roque Rodríguez · A. Cortés · T. Margalef · Computer Science, Environmental Science ·

2010 10th IEEE/ACM International Conference on… · 17 May 2010

TLDR This paper proposes a strategy for data injection in distributed environments and aims to build a system that dynamically adapts to constant changes in environmental conditions when a hazard occurs and under strict real-time deadlines. Expand

5 IEEE Save Cite

A review: antimicrobial resistance data mining models and prediction methods study for pathogenic bacteria

Xinxing Li · Ziyi Zhang · Buwen Liang · Fei Ye · Weiwei Gong · Medicine, Biology ·

Journal of antibiotics (Tokyo. 1968) · 14 September 2021

TLDR The current research results on antimicrobials and the development trend are summarized to provide a systematic and comprehensive reference for the research on antimicrobial resistance. Expand

13 Springer Nature Save Cite

A systematic review of dengue outbreak prediction models: Current scenario and future directions

Xing Yu Leung · R. Islam · +6 authors · M. N. Karim · Medicine, Environmental Science · medRxiv ·

8 July 2022

TLDR A systematic **review** of the published literature that used quantitative models to predict dengue outbreaks and collates plausible predictors and methodological approaches, which will contribute to robust modelling in diverse settings and populations. Expand

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